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THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF ELECTRICAL ENGINEERING AND
TELECOMMUNICATIONS

ELEC4631
Continuous-Time Control System Design

Final Examination

T2 2022

- Time Allowed: 2 hours 5 minutes (exam time) plus 30 minutes (scanning and uploading solutions)
- This examination is open-book. You may use course notes and other course materials.
- NO collaborations allowed and the work submitted must be your own. During the exam you may NOT be in contact with anyone else via any form of communication media (email, messaging, phone, video conferencing, etc). Violating this will be considered an academic misconduct and disciplinary action will be taken against anyone who is proven to have violated this rule.
- This paper contains this page and 6 other pages containing four (4) Questions. There is a total of 7 pages including this page.
- Total marks available: 100.
- This examination is worth 56% of the total course mark.
- All questions are NOT of EQUAL value. Answer ALL questions.
- You may use a calculator or computer software such as MATLAB only for real and complex arithmetic, evaluating powers (squares, cubes, etc), roots (square roots, cube roots, etc) and the standard functions e^x , 10^x , $\ln(x)$, $\log_{10}(x)$, $\sin(x)$, $\cos(x)$ and $\tan(x)$ for real numbers x , and for the addition, multiplication and inversion of matrices. You may not use Matlab or any other software for symbolic manipulations. ALL workings must be shown.
- All answers must be written in ink. Except where they are expressly required, pencils may be used only for drawing, sketching or graphical work.
- Uploaded scans must be readable, clear and of appropriate size. That is, they must be fit for marking. Questions may not be marked and will receive zero marks if they are not legible.
- Scan and upload your solutions to all problems together in a single pdf file.

QUESTION 1

(25 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

(a) Consider a third order LTI system $\dot{x} = Ax$ with

$$A = \begin{bmatrix} -(n_3 + 2) & -(n_5 + 1) & 0 \\ n_4 + 3 & n_3 + 2 & -1 \\ 0 & (n_3 + 2)^2 & -(n_3 + 2) \end{bmatrix}.$$

Answer the following questions:

(i) Calculate the characteristic polynomial and the eigenvalues of A . (10 marks)

(ii) Determine the stability property of the system (asymptotically stable, marginally stable or unstable) and justify your answer. (5 marks)

(b) Consider the second order LTI state space system

$$S = \left(\begin{bmatrix} -(n_2 + n_4)/2 & 5 \\ 20 & -(n_2 + n_4)/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \end{bmatrix}, 0 \right)$$

Answer the following questions:

(i) Calculate the transfer function of the system. Keep all coefficients in exact rational form, do not do any rounding. (5 marks)

(ii) Determine if the system is BIBO stable and justify your answer. (5 marks)

⁰Please turn over - Question 2 on next page.

QUESTION 2

(32 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

(a) Consider the SISO LTI state-space system,

$$S = \left(\begin{bmatrix} 0 & n_5 + 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 \\ n_1 + 2 \end{bmatrix}, \begin{bmatrix} n_3 + 3 & 0 \end{bmatrix}, 0 \right),$$

with the state $x(t) = (x_1(t), x_2(t))^T$ and the infinite-horizon LQ performance index

$$J_\infty(x_0, u) = \int_0^\infty (16x_1(t)^2 + u(t)^2) dt.$$

Answer the following questions:

- (i) Verify that (A, B) is a controllable pair and $(Q^{1/2}, A)$ is an observable pair. (6 marks)
- (ii) Solve the CARE equation for P and determine the LQR control law u_{opt} . (10 marks)
- (iii) Determine the optimal LQ cost $J_{\infty, \text{opt}}(x_0)$ when the initial state is $x_0 = (1, 1)^T$. (3 marks)

(b) Let $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times 1}$, $Q \in \text{sym}(\mathbb{R}^{n \times n})$ with $Q > 0$, and $r > 0$ a positive scalar. Consider the order n system

$$\dot{x} = Ax + Bu,$$

with u a scalar control signal, and suppose that the CARE equation:

$$A^T P + PA - r^{-1} P B B^T P + Q = 0,$$

has a solution $P \in \text{sym}(\mathbb{R}^{n \times n})$ with $P > 0$. Show that the linear state feedback control law $u = -r^{-1} B^T P x$ stabilises the closed-loop system. That is, show that the eigenvalues of $A_{\text{cl}} = A - r^{-1} B B^T P$ must all be in the LHP.

(5 marks)

(c) Consider the controlled ODE in $S = \mathbb{R}^2$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -(n_1 + 1)x_1 + (n_2 + 1)x_2 + (n_3 + 1)x_1 x_2 \\ -(n_2 + 1)x_1 - (n_4 + 1)x_2 + (n_5 + 1)x_1^2 + (n_6 + 1)x_2^2 + u \end{bmatrix},$$

and the candidate Lyapunov function $V(x_1, x_2) = (1/2)(x_1^2 + x_2^2)$.

⁰Please turn over - Question 2 continued on next page.

Answer the following questions:

- (i) Compute $\dot{V}(x_1, x_2)$.(4 marks)
- (ii) Show that there is a control law u such that $\dot{V}(x_1, x_2)$ is a negative definite quadratic form. Thus this control law renders the origin globally asymptotically stable.(4 marks)

⁰Please turn over - Question 3 on next page.

QUESTION 3

(15 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

Answer the following questions:

(a) Consider the following LMI in the decision variable $x = (x_1, x_2)$:

$$\begin{bmatrix} x_1 & (n_1 + 1)x_1 - (n_3 + 2)x_2 \\ (n_1 + 1)x_1 - (n_3 + 2)x_2 & x_1 \end{bmatrix} > 0.$$

Sketch the region in $\mathbb{R}^2$ where the LMI has a solution, if the LMI is strictly feasible.

(10 marks)

(b) Let $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ be given non-zero constant matrices. Consider the following matrix inequalities in the matrix variables $X \in \text{sym}(\mathbb{R}^{n \times n})$ and $Y \in \mathbb{R}^{n \times n}$,

$$\begin{aligned} X &> 0 \\ -XA^TX^{-1}Y - Y^TX^{-1}AX + YX^{-1}Y^T - XBB^TX &< 0. \end{aligned}$$

Show that the matrix inequalities above can be expressed as an LMI in the new matrix variables $Z = X^{-1}$ and $W = X^{-1}YX^{-1}$.

(5 marks)

⁰Please turn over - Question 4 on next page.

QUESTION 4

(28 marks)

In this question, numerical values $n_1, n_2, \dots, n_7$ must be substituted from your ZID number $zn_7n_6n_5n_4n_3n_2n_1$. For example, if your ZID is **z5412079** then make the substitutions $n_1 = 9, n_2 = 7, n_3 = 0, n_4 = 2, n_5 = 1, n_6 = 4$ and $n_7 = 5$. Write down values of $n_1, n_2, \dots, n_7$ on your exam paper.

ALL workings must be included in your solution. No workings, no marks.

Consider the SISO LTI system

$$S = \left(\begin{bmatrix} 0 & n_5 + 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 \\ n_1 + 2 \end{bmatrix}, \begin{bmatrix} n_3 + 3 & 0 \end{bmatrix}, 0 \right).$$

From Lecture 11, you can design an output feedback controller with zero assignment estimator Z (with observer state $\xi^o(t)$) for S so that the controlled system can track any step input $r(t)$ with zero error in steady state, as shown in Figure 1. M is a 2×1 vector, N is a non-zero scalar, K is the 1×2 state-feedback gain, and L is the 2×1 observer gain.

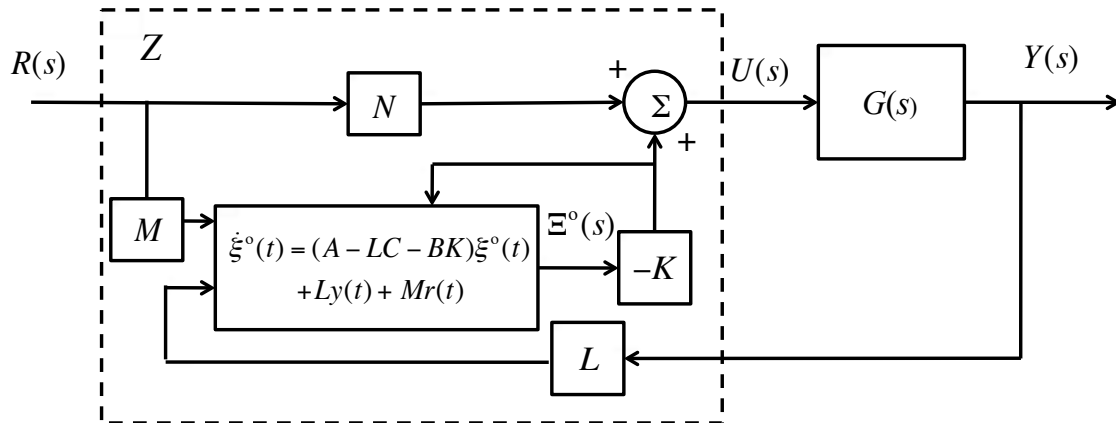


Figure 1: The system S , represented by its transfer function $G(s)$, with an output feedback controller with zero assignment estimator Z .

Answer the following questions:

- (a) Using Ackermann's formula, design K such that two eigenvalues of the closed-loop system are at $p_1 = -4 + 3j$ and $p_2 = -4 - 3j$. Here $j = \sqrt{-1}$. Show the calculations for $R(A)$, $\mathcal{C}$ and K .

(8 marks)

- (b) Using Ackermann's formula, design L such that the observer contributes closed-loop eigenvalues at $p_3 = -0.4 + 0.05j$ and $p_4 = -0.4 - 0.05j$. Show the calculations for $R(A)$, $\mathcal{O}$ and L .

(8 marks)

⁰Please turn over - Question 4 continued on next page.

- (c) Using Ackermann's formula, design β so that two of the closed-loop zeros are at $z_1 = -0.45$ and $z_2 = -0.35$. Show all calculations.

(8 marks)

- (d) Determine the closed-loop transfer function $H_{cl}(s)$ from the reference signal $R(s)$ to the output $Y(s)$.

(4 marks)

⁰End of paper.